

MTH-100

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Question No 01 :

To find the composite function where:

- $f(x) = \sqrt{x}$
- $g(x) = \frac{1}{x}$
- $h(x) = x^2$

we will calculate it step by step.

Step 1: Find $h(x)$

First, we start with $h(x)$:

$$H(x) = x^2$$

Step 2: Find $g(h(x))$

Now, substitute $h(x)$ into $g(x)$:

$$G(h(x)) = g(x^2) = \frac{1}{x^2}$$

Step 3: Find $f(g(h(x)))$

Now, substitute $g(h(x))$ into $f(x)$:

$$f(g(h(x))) = f\left(\frac{1}{x^2}\right) = \sqrt{\frac{1}{x^2}} = \frac{1}{|x|}$$

Final Result

Thus, the composite function $(f \circ g \circ h)$ is:

$$(f \circ g \circ h)(x) = \frac{1}{|x|}$$

Question no 2 :

Determine the domain and range of fog if $f(x) = 1/x$ and $g(x) = x + 1$.

Answer

To determine the domain and range of the composite function $(f \circ g)$, where

- $f(x) = \frac{1}{x}$
- $g(x) = x + 1$

we will first compute $(f \circ g)(x)$ and then analyze its domain and range.

Step 1: Find $(f \circ g)(x)$

The composite function is defined as:

$$(f \circ g)(x) = f(g(x)) = f(x+1)$$

Substituting $g(x)$ into $f(x)$

$$F(x+1) = \frac{1}{x+1}$$

Step 2: Determine the Domain

The domain of $(f \circ g)(x) = \frac{1}{x+1}$ consists of all x values for which the function is defined. The function is undefined where the denominator is zero:

$$x+1 \neq 0 \rightarrow x \neq -1$$

Thus, the domain is:

Domain: $(-\infty, -1) \cup (-1, \infty)$

Step 3: Determine the Range

The range of $f(x) = \frac{1}{x}$ is all real numbers except zero. Since $g(x)$ shifts the input but does not affect the overall behavior of f beyond avoiding division by zero, the range of $(f \circ g)(x) = \frac{1}{x+1}$ remains the same.

As x approaches -1 from the left, $\frac{1}{x+1}$ approaches $-\infty$, and as x approaches -1 from the right, $\frac{1}{x+1}$ approaches $+\infty$. Therefore, the function will never equal zero.

Thus, the range is:

Range : $(-\infty, 0) \cup (0, \infty)$

Question: 3

Solve the system of linear equations:

$$3x + 4y - 7z = 2$$

$$-2x + y - 2z = -6$$

$$x + y + z = 2$$

Simplify each equations:

Equation 1 $3x + 4y - 7z = 2$

Equation 2 $-2x + y - 2z = -6$

Equation 3 $x + y + z = 2$

Step 2 Solve for Y in term of X equation 2

From eq 2

$$Y = 2x - 4$$

Step 3 : Substitute $y = 2x - 4$ into equation 1

$$3x + 4(2x - 4) = 74$$

$$3x + 8x - 16 = 74$$

$$11x = 90$$

$$X = \frac{90}{11}$$

Step 4 : Substitute $x = \frac{90}{11}$ into $y = 2x - 4$ to find y

$$Y = 2\left(\frac{90}{11}\right) - 4$$

$$Y = \frac{180}{11} - \frac{44}{11}$$

$$Y = \frac{136}{11}$$

Step 5 Substitute $x = \frac{90}{11}$ and $y = \frac{136}{11}$ into eq 3 to find Z

$$X + y + z = 2$$

$$\frac{90}{11} + \frac{136}{11} + z = 2$$

$$\frac{226}{11} + z = 2$$

$$Z = 2 - \frac{226}{11}$$

$$Z = \frac{22 - 226}{11}$$

$$Z = \frac{-204}{11}$$

Question No 04 :

Solve the quadratic equation $y = x^2 + 3x + 1$.

Answer

Use Quadratic Formula

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2ac}$$

$$a = 1, b = 3, c = 1$$

Put all values in above formula:

$$X = \frac{-3 \pm \sqrt{(3)^2 - 4(1)(1)}}{2(1)(1)}$$

$$= \frac{-3 \pm \sqrt{9 - 4}}{2}$$

$$= \frac{-3 \pm \sqrt{5}}{2}$$

The solutions to the Equations are:

$$X = \frac{-3 - \sqrt{5}}{2}, \quad x = \frac{-3 + \sqrt{5}}{2}$$

